CHAPTER 1

Getting Started

We shall start at the very beginning. PHP is an interpreted language, which means that every time the source code of a PHP program runs, it is parsed (broken into pieces and analyzed) and executed by another program called the interpreter. The PHP interpreter is normally either a standalone binary executable or shared library embedded into a web server (Apache), and before we can run any PHP code, we need to make sure the interpreter is available on the system.

The standalone PHP binary is designed to be invoked like any command-line binaries on a system; you can type its name into a command shell (such as the popular Bash: https://www.gnu.org/software/bash/). It can be given a filename argument when invoked ("arguments" are the characters written after the command) and that file will then be interpreted and executed. It can also run interactively (individual statements are typed in and executed as they are entered), or even run in a special "cli-server" mode where the interpreter binds to a network port and offers basic HTTP functionality.

When running via a "full-fat" web server the PHP interpreter can either be a compiled module linked directly to the web server binary (e.g., a Windows `.dll` or Linux `.so` library file) or a CGI processor (which is a separate executable binary that the web server calls via the Common Gateway Interface). Don't panic if any of this is unfamiliar; all will be explained.

Many operating systems come with PHP preinstalled or at least have it as an optional package during the installation process. You can check if PHP is available on a Linux-based machine (or BSD-based, or actual Unix… these types of operating system are often referred to as *nix) with a simple shell command:

```
$ which php
```

```
Output:
/usr/local/bin/php
```


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